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Framing climate risk: the role of local and international media in reporting on Japan's omiwatari ice tradition

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Abstract

This study examines how local media in Japan report on the risks of climate change through the lens of a cultural tradition, focusing on the Shinto ritual of omiwatari in Nagano Prefecture. *Omiwatari* celebrates ice ridges on Lake Suwa that are believed to be the footprints of a local deity, but are increasingly threatened by warming winters. By analyzing online news articles from Japanese and international media, supplemented by fieldwork and interviews with local journalists, the research highlights contrasting media frames. International media emphasize the disappearance of the ice ridges as a symbol of global climate change, aiming to promote broader awareness and mitigation efforts. In contrast, Japanese national and local media focus their coverage on the observations of the shrine community and predict the formation of the ridges without framing these changes in the context of climate change. Despite this divergence, the resilience of *omiwatari* storytelling offers opportunities to adapt the tradition to changing environmental challenges, preserving cultural continuity even as its iconic ice ridges fade.

Keywords: Intangible cultural heritage, Media, Journalistic norms, Climate change adaptation, Loss and damage, Japanese winter tradition

Introduction

Climate change is profoundly altering the natural environment, causing a decline in snow and ice that threatens Japanese winter cultural heritage and its intergenerational transmission. In this context, media can play a pivotal role in raising public awareness of climate risks and fostering community engagement. This study examines how winter cultural heritage is represented in Japanese local and national media, with a focus on a local Shinto practice in winter. It also compares these frames with international media coverage to examine how media framing can contribute to preserving intangible cultural heritage in the face of climate change.

This study examines the case of *omiwatari*, a traditional Shinto practice associated with the ice formations in Nagano Prefecture. Lake Suwa is known for producing significant ice ridges during harsh winter conditions. According to local legend, this mysterious phenomenon is attributed to a path traversed by a male deity across the

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lake. Yatsurugi shrine in the Suwa area has historically offered blessings related to the occurrence of these ice ridges. However, due to recent warming winters, the ice ridges have not formed for six consecutive years as of February 2024. Nevertheless, the priest and parishioners of the shrine remain hopeful and are patiently awaiting the return of the *omiwatari* phenomenon each winter.

Intangible cultural heritage, such as the *omiwatari*, is recognized when a community shares a sense of place or identity and sees itself as part of that heritage (Blake 2008). This heritage is passed down through continuous recreation by the community (UNESCO 2009). Strong community ties, also known as social capital (Putnam 2000), are essential for the transmission of intangible cultural heritage (Blake 2008; UNESCO 2009). In northern Japan, communities have developed social capital through the ongoing practice of winter traditions, which are vital for their survival during the winter months (Fraser 2021; Hikichi et al. 2020). Consequently, climate change poses a threat to the survival of these local winter traditions, potentially undermining the social relationships and capital that these communities have built around their cultural heritage.

The significance of media coverage related to climate change is steadily increasing (Barkemeyer et al. 2017; Bolsen & Shapiro 2018; Howarth & Anderson 2019). Media coverage of climate change varies by country and is influenced by each nation's political, cultural and other contexts (Anderson 2009; Barkemeyer et al. 2017; Schmidt et al. 2013). For instance, in the Czech Republic, discussions on coal mining are often detached from climate change due to the country's reliance on coal (Lehotský et al. 2019). Despite these significant concerns, public interest in climate change remains relatively low in Japan (Barkemeyer et al. 2017), especially among younger generations (Hasegawa 2020). This contrasts with their counterparts in other countries and suggests a unique approach to climate change within Japanese civil society.

While the effects of climate change are often recognized locally, research and policies are primarily made at the global or national levels (Kleinberga 2022; Spradlin & Givens 2022). This can create a disconnect between the information provided by national media and what local governments and communities need for climate preparedness. Local media play an important role in addressing this gap (Howarth & Anderson 2019, p. 715). In the UK, for instance, local newspapers frame climate change as a local issue requiring local solutions (Brown et al. 2011). In Japan, after the Great East Japan Earthquake, local newspapers focused not only on the overall damage but also on detailed information vital for community rebuilding (Numada & Meguro 2013). This highlights the influence of local media on disaster risk management.

The survival of winter intangible cultural heritage relies heavily on snow and ice, both of which are increasingly threatened by climate change. Therefore, mitigating climate change is essential to preserving this heritage (Carvalho Resende et al. 2022). However, climate science indicates that even if greenhouse gas emissions were to be completely eliminated today, the effects of climate change would still persist for some time (IPCC 2022). Currently, the risks of loss and damage to winter intangible cultural heritage are adversely affecting the livelihoods, mental health, and overall well-being of those who have long safeguarding these traditions (IPCC 2019; Jurt et al. 2015; Middleton et al. 2020; Vuille et al. 2018).

Intangible cultural heritage is considered a "living culture" because it is continuously passed down and recreated within society and its environment, reflecting its significance and value in its environment (Lenzerini 2011). While the physical impacts of climate change may impose limitations on adaptation efforts, it remains possible to transcend social boundaries by transforming the values, perspectives, and assumptions of individuals and communities (Adger et al. 2008).

Media coverage of climate change has largely focused on economic and human losses (McHugh et al. 2021). Additionally, there is a lack of research on climate change as it pertains to intangible cultural heritage worldwide (Orr et al. 2021). In Japan, newspaper coverage significantly influences public awareness of climate change (Sampei & Aoyagi-Usui 2009), but the focus tends to be on climate science, policy, and international negotiations (Asayama & Ishii 2017; Konishi 2017). Local cultural issues, such as the *omiwatari*, which are of significant interest to local communities, often receive insufficient attention, even in the national climate impact assessment reports published in 2015[1] and 2020[2]. Given the rapid pace of global warming, it is crucial to analyze how the media frames the risks to the Japan's winter tradition. This analysis will shed light on the role of media in shaping effective climate risk management strategies for preserving these cultural traditions.

This study seeks to address the following research questions are as follows:

> RQ1: How do Japanese and international media frame the omiwatari, and how do these framings differ across media types?
> RQ2: What themes and topics, particulary related to climate change and cultural heritage, are emphasized in media coverage of the omiwatari?

To achieve this, the research employs a mixed-methods approach, combining content analysis of newspaper articles from both Japan and other countries, ethnographic fieldwork, and interviews with journalists covering the *omiwatari* tradition. This comprehensive methodology aims to uncover the journalistic norms that influence how the media frames the *omiwatari* tradition within the context of climate change.

The following sections outline the theoretical framework of the study, detail the *omiwatari* case being examined, and describe the methods used. Afterward, the results are presented, followed by discussions on how Japanese media coverage reflects the diminishing significance of the omiwatari due to climate change.

Theoretical framewok

Media framing and journalistic norms play a crucial role in shaping public understanding of environmental and climate risks, helping communities to identify challenges and raising awareness of potential solutions (Boykoff & Boykoff 2007; Hase et al. 2021; Miller & Riechert 2013). However, climate change reporting often struggles to sustain consistent news coverage, leading to a tendency for short-term public attention (Schmidt

[1] Central Environment Council. (2015). Report on Assessment of Impacts of Climate Change in Japan and Future Challenges (Comment Submission). https://www.env.go.jp/content/900451428.pdf. (Accessed on November 22, 2024)

[2] Ministry of the Environment. (2020). Assessment Report on limate Change Impacts in Japan. https://www.env.go.jp/content/000047546.pdf. (Accessed on November 22, 2024).

et al. 2013; Wozniak et al. 2021). Additionally, the norm of balanced reporting in climate journalism, where anthropogenic and natural causes of climate change are given equal weight, have delayed significant climate action particularly in countries like the United States (Anderson 2017; Boykoff & Boykoff 2004). Recognizing these challenges, recent studies have investigated media content that promote effective climate action, often employing content analysis methodologies (e.g., Hase et al. 2021; Howarth & Anderson 2019; Wozniak et al. 2015).

The rise of online media has further expanded the scale and diversity of climate change reporting, incorporating topics beyond science and the environment, such as business, energy, sports, and culture (Schäfer & Painter 2021). These shifts necessitate analyzing media coverage from the perspectives of journalists, whose choices and interpretations significantlly influence the narratives presented to the public (Schäfer & Painter 2021).

Building on these principles, this study to examines how media framing and journalistic norms shape the coverage of winter cultural heritage under climate change. Media framing organizes public understanding by embedding issues within specific narratives (Guenther et al. 2024; Stecula & Merkley 2019). While journalistic norms are intended to promote fair and balanced reporting, they can unintentionally lead to biased or incomplete coverage if not critically applied.

Type of framing

Iyengar (1994) examines the impact of framing on public opinion in political communication, distinguishing between two primary types: thematic framing and episodic framing.

a. Thematic Framing: This approach places public issues within broader, abstract contexts, focusing on general trends, systemic causes, and long-term outcomes. Thematic framing often provides a comprehensive understanding of complex issues. For instance, in disaster reporting, it might explore land-use policies or social dynamics that contribute to vulnerability rather than focusing on individual events.
b. Episodic Framing: This type of framing focuses on specific incidents or case studies, often highlighting individual stories or unique events. For instance, disaster coverage might center on the struggles of a single family, drawing attention to their immediate experiences while overlooking broader systemic causes. Episodic framing influences audience perceptions of responsibility by emphasizing personal or localized factors.

These framing styles significantly affect how audiences perceive responsibility for climate issues. While thematic framing often encourages systemic solutions, episodic framing may limit discussions to immediate, localized actions.

Journalistic norms in climate change reporting

Journalistic norms influence how climate change is framed, selected, and presented in media coverage, shaping public understanding and potentially hindering effective action. Boykoff & Boykoff, (2007) identify key norms, such as (1) personalization, which emphasizes human interest over systemic issues; (2) dramatization, which prioritizes sensational events over continuity; (3) novelty, which favors fresh angles while sidelining

ongoing narratives; and (4) authority-order, which leans heavily on official sources. van Eck et al., (2019) extend this analysis to digital platforms, showing how norms like personalization are adapted by bloggers to broaden engagement through personal anecdotes. Despite these adaptations, entrenched norms across media types often prioritize episodic framing and short-term crises, limiting the contextualization of long-term climate risks and collective solution (van Eck et al. 2019).

The first and last norms are particularly evident in climate change reporting. Personalization often emphasizes individual figures, such as young climate activist Greta Thunberg, sometimes diverting attention from broader systemic issues related to anthropogenic climate change (Boykoff & Boykoff 2007; Jacobsson 2021). Similarly, authority-order norms reflect journalists' reliance on experts and officials, such as the Intergovernmental Panel on Climate Change (IPCC), to dominate narratives to reassure the public (Boykoff & Boykoff 2007; van Eck et al. 2019). In climate change discourse, these norms shape how issues are framed and influence the prioritization of topics. For instance, an emphasis on immediate risks often overshadows gradual, systemic processes, such as the cultural impacts of environmental changes.

Case study: the *omiwatari* ice tradition of Nagano, Japan.

Located in Nagano Prefecture, Suwa is a mountainous region with a population of around 198,000. Lake Suwa, nestled among the Northern Alps, freezes in winter, creating striking ice ridges accompanied by massive echoing crackind sounds. Local tradition holds that these ice ridges, believed to be the footsteps of a male deity crossing the lake to visit his wife on the opposite shore, symbolize the awe-inspiring power of nature (Nakazawa 1965). Historically, upon hearing these sounds, local residents would gather to observe the ice ridges believed to represent the passage of the god (*omiwatari*) and perform rituals (Ito 1957). Due to this mysterious scenery, the cultural significance of the *omiwatari* phenomenon has been widely acknowledged in ukiyo-e prints and kabuki plays.

Traditionally, when the *omiwatari* phenomenon occurs, Yatsurugi Shrine in Suwa holds a blessing ceremony and reports the event to the local deities enshrined at Suwa Taisha shrine, as well as to the national authorities at that time. A record of this report is kept in an archive known as the '*Mitarai-cho*' (literally 'crossing record') (Ito 1957). In years when the *omiwatari* phenomenon does not occur, it is referred to as '*ake-no-umi*' (literally 'the sea of dawn'), and no Shinto rituals are performed on the ice. However, the reporting to the relevant organizations as well as the documentation of the ice record are still carried out. In the Suwa region, this set of traditional practices is collectively known as the *omiwatari* and ice on the lake has been observed for over five centuries (Saito 2019). Due to its uniqueness as a geographical phenomenon, the occurrence of the *omiwatari* in 2018 was reported by the national media (I. Miyasaka 2018; Yorimitsu 2018).

In recent years, however, climate change has threatened the occurence of the *omiwatari* phenomenon (See Fig. 1). Arai (2000) found that the frequency of *ake-no-umi* has increased since the Industrial Revolution in the nineteenth century. The continuity of *ake-no-umi* has transformed the ritual practice. In the past, local people would summon the priest of Yatsurugi Shrine to perform the ritual when the *omiwatari* phenomenon occurred. Concerned about preserving the tradition amidst consecutive years without

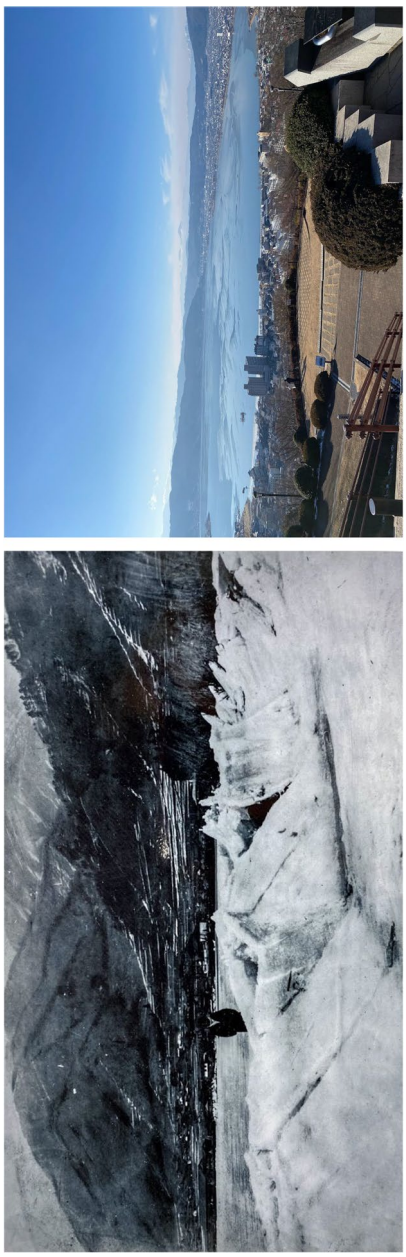


Fig. 1 The changes in Lake Suwa over the past century

the *omiwatari* ritual on ice, the current priest, Kiyoshi Miyasaka, introduced daily morning observation sessions of Lake Suwa when he took over the role from his father (Nagano Nippo 2022). This early morning observation sessions are conducted with the shrine parishioners to predict the possibility of the *omiwatari* phenomenon occurring. The observations take place annually from early January to early February, in accordance with the traditional Japanese calendar's Twenty-Four Solar Terms. This practice has become a seasonal tradition, and the local newspaper reports on it almost daily throughout the observation period.

The parishioners of Yatsurugi Shrine have documented the *omiwatari* phenomenon in the *miwatari-cho*, the ice-observation archive that has been preserved for 582 years, dating back to 1443. Despite its cultural significance, modernization disrupted this tradition with the abolition of fudal system, which removed responsibility for Yatsurugi shrine to record the ice formation (M. Miyasaka 2007). Recognizing the scientific value of these long-term ice records, meteorologist Sakuhei Fujiwhara worked to recover missing data, and the tradition was revived (Fig. 2). Today, the shrine continues to report the presence or absence of the *omiwatari* phenomenon annually to the Japan Meteorological Agency and the Imperial Household Agency.

Dr. Fujiwhara's contributions are highly regarded in current climate change research. The records of the *omiwatari* phenomenon offer important data for paleoclimatologists, including Dr. Magnuson and others, in tracing changes in the climate (Knoll et al. 2019; Mikami 2008, 2023; Mikami & Hirano 2018; Sharma et al. 2019, 2020).

In 2024, Yatsurugi Shrine declared the 78th *ake-no umi* since 1443, marking the sixth consecutive absence of the *omiwatari* phenomenon since 2018. The Nagano Meteorological Observatory (2022) projects a temperature increases of 2.0 °C to 4.6 °C in Nagano Prefecture, depending on the severity of CO2 emission scenarios.

This side-by-side comparison showcases two winter scenes of Lake Suwa, separated by a century. On the left, the 1913 image captures towering ice ridges that reach heights comparable to an adult standing nearby. This scene illustrates the cultural significance of the *omiwatari* tradition, as local communities depended on these ice formations for their seasonal activities and celebrations. On the right, the 2023 image presents a stark contrast, depicting Lake Suwa with a completely ice-free surface. This contemporary view highlights the profound impacts of climate change, revealing the diminishing presence of traditional winter ice formations that have historically shaped the region's

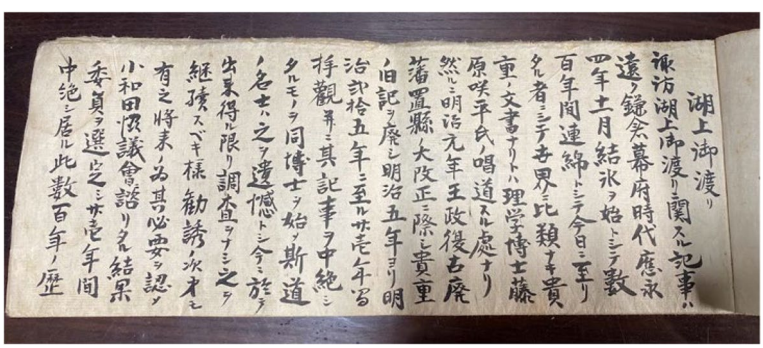

諏訪湖上御渡リ顕ハル記事ハ
遠ク鎌倉幕府時代ヨリ永
享十七年十月結氷ヲ始トシテ数
百年間連綿トシテ今日ニ至リ
タル者ニシテ各国ニ比類ナキ貴
重ノ文書ナリトハ理学博士藤
原咲平氏ノ唱道スル処ナリ
然ルニ明治元年王政復古ノ
藩置縣ノ大改正ノ際ニ貴重
ノ旧記ヲ廃シ明治五年ヨリ明
治卅五年ニ至ルマデノ年々
将観等ノ其記事ヲ中絶シ
タルモノヲ同博士ヲ始メ斯道
ノ名士ハ之ヲ遺憾トシ今ニ於テ
ハ之ヲ得ル限リ調査ヲナシ之ヲ
経緯スベキ様勧誘ノ次第モ
有之将来ノ為メ必要ヲ認メ
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委員ヲ選ビ之ヲ永ク年間
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Fig. 2 An introductory page to the ice observation records that are filling in the missing data gap

cultural and environmental landscape. (Photo by the author; left part used with permission from Yatsurugi Shrine).

This document outlines the background regarding the suspension and resumption of the record of ice cracks on Lake Suwa. “Dr. Sakuhei Fujiwhara, a doctor of science, emphasized the importance of the continuous record of ice ridges on Lake Suwa, which has been maintained for several centuries. He considered this record to be an invaluable and unique document globally. Disappointed by the discontinuation of the observation following the abolition of the feudal system, he conducted extensive research in collaboration with local experts to fill the gap in missing data and advocated for the continuation of the documentation of ice ridges.” (Photo by the author, used with permission from Yatsurugi Shrine)

Method

This research was conducted between 2021 and 2023, a period chosen due to the global COVID-19 pandemic. During this period, media coverage of social and economic events had significantly decreased. It was anticipated that media attention would likely shift toward cultural topics during this time, making it an opportune period for the study. The research employed content analysis, focusing on online articles from both overseas and Japan that reported on the *omiwatari*. The first step in the analysis involved collecting articles on the topic, which formed the primary data for the study.

There remain ongoing challenges regarding the use of online articles as research data, such as inconsistencies in online databases and the lack of a well-established methodology for their collection (Blatchford 2019). However, given the global trend of increasing subscriptions to online articles in contrast to the declining readership of traditional newspapers, online news articles are becoming increasingly valuable as sources of information (Goyanes et al. 2021). For these reasons, online articles were judged appropriate as the subject of this research.

The comparative content analysis examined international and Japanese online articles reporting on the *omiwatari*. International articles were collected via Google as it emphasizes its global reach by offering access to information worldwide (Mihelj & Jiménez-Martínez 2021; van Dijck et al. 2018). Using the keywords "*omiwatari*," "lake," and "suwa," fourteen English-language articles were collected through the platform, including one originally written in Dutch. The Dutch article was translated into English using machine translation for consistent analysis. Due to the limited availability of foreign articles, publication dates were not restricted for non-Japanese sources.

Japanese articles were taken from national and local newspapers published online between December 2021 and February 2022, the period when the ice ridge was most likely to have occurred. A total of 105 Japanese articles were collected from the regional sections of three national newspapers, three regional newspapers, one Japanese news agency, and three television station websites.

The comparative content analysis examined Japanese and international online news articles that reported on the *omiwatari*. The analysis was further complemented by observations and focus group interviews with local journalists in Suwa who wrote the articles.

The collected articles were systematically analyzed by categorizing it into international, national, and local media. To achieve this, all articles were read multiple times by the author, who identified categories and codes to serve as the basis for analysis. Definitions for each category and code were inductively established (Bingham 2023). A codebook was developed to ensure consistent classification of the data.

To complement the content analysis, ethnographic fieldwork was structured around participant observation and a focus group interview with reporters at observation events, conducted to capture their perspectives and experiences. Observations took place annually from 2021 to 2023 during January and February, lasting about one week each year. Each year, approximately 10 reporters attended the observation event. In 2022, semi-structured a focus group interview were conducted with five journalists following an observation event. One journalist represented a national media outlet, while the others were from local media, including the Suwa branch of national newspaper companies. A separate individual interview was conducted with a local journalist using the same set of questions. This structured approach facilitated systematic data collection among journalists with varying backgrounds. In total, data were collected from six journalits. The following questions were asked:

> How often do you join the morning observation, and why?
> Are your articles published every day?
> What is the focal point of your reporting?
> Have you ever reported on the *omiwatari* from a climate change perspective?
> How does your company prioritize climate change issues (low, medium, or high)?

The interview data were coded according to the participants' experiences and perspectives on the *omiwatari* observation, and the themes and stories they focused on were analyzed in conjunction with the results of the content analysis. This helped develop a narrative linking journalistic practices with the framing of the *omiwatari* in the media.

This study has some limitations. Although the sample size of this study is small, its qualitative, exploratory approach enabled it to capture nuanced, context-specific insights into the under-researched intersection of media coverage and intangible cultural heritage. Though focused on a single region, the study offers a framework applicable to similar contexts globally. Lastly, while the perspectives of newsroom decision-makers, such as editors, were not included, the direct insights from local journalists offer valuable understandings of how media practices shape the coverage of climate risks to the cultural heritage.

Results

Focal point of foreign articles

The results of coding the online content from overseas are presented in Table 1. It is evident that all of the articles, except for one from a natural ecology journal, were scientific stories focused on climate change that attributed to the Industrial Revolution. This scientific perspective was influenced by the research of Dr. Magnuson. Nearly all of these articles connected the phenomenon of ice ridges to local myths and highlighted that the priest of Yatsurugi shrine maintains the archive of long-term ice records. Among the 14

Table 1 Major topics discussed about the *omiwatari* Foreign media clearly frame the continuous absence of the *omiwatari* ice ridges in the context of climate change. On the contrary, Japanese media seldom mention climate change in their articles. In the table, different media belonging to the same category are designated by alphabetical labels for clear identification

	Foreign	TV	TV	TV	Wire	National Paper			Local Paper		
	media	a	b	c	sevice	a	b	c	a	b	c
Number of articles / Topic	14	1	1	2	1	11	8	9	14	35	24
Climate change	13	0	0	0	0	0	1	0	0	2	1
Industrial Revolution	13	0	0	0	0	0	0	0	0	0	0
Ice cover	10	1	1	2	1	9	5	3	9	28	20
Ice ridges	12	0	0	2	1	7	7	7	8	23	5
Morning observation	2	1	1	2	1	9	6	5	9	32	20
Visitors	0	0	0	0	1	0	2	1	1	5	3
Measuring temperatures	1	0	0	2	0	7	3	6	8	26	18
Shinto myths	13	0	0	0	0	0	0	0	1	1	1
Long tradition	14	0	0	1	0	4	4	1	2	1	1
Omiwatari ice records	14	0	0	1	0	5	4	1	2	2	1
Priest	14	0	0	2	1	10	6	9	12	29	16
Parishioners	0	0	0	2	0	8	4	2	5	7	15
Local culture	9	0	0	1	0	6	0	2	4	3	8
Ecology	9	0	0	0	0	1	0	0	0	1	7
No longer ice ridges	5	0	0	0	0	0	0	0	0	0	0
Continuous absence	14	0	0	2	1	4	4	1	4	2	2
Expectation	2	1	1	1	0	6	3	1	7	10	13
Ake-no-umi	2	0	0	0	1	2	1	2	2	2	8
Cases other than omiwatari											
Lakes in Japan	0	0	0	0	0	0	1	0	0	0	0
Torne River, Finland	8	0	0	0	0	0	0	0	0	0	0
North American Great lakes	2	0	0	0	0	0	0	0	0	0	0
Henry David Thoreau	2	0	0	0	0	0	0	0	0	0	0
North Atlantic Oscillation	1	0	0	0	0	0	0	0	0	0	0
Lake Constance in Europe	1	0	0	0	0	0	0	0	0	0	0
Indonesia	1	0	0	0	0	0	0	0	0	0	0
Native Americans	1	0	0	0	0	0	0	0	0	0	0
Indigenous knowledge	1	0	0	0	0	0	0	0	0	0	0
Scientifice topics	13	0	0	0	0	0	2	0	0	4	1
Scientists											
Magnuson	10	0	0	0	0	0	0	0	0	0	0
Sharma	2	0	0	0	0	0	0	0	0	0	0
Japanese scientists	4	0	0	0	0	0	1	0	0	1	2

articles analyzed, all mentioned the absence of long-term ice ridges. Five of them concluded that the phenomenon would not reappear in the future, nor would the male deity cross Lake Suwa. One article specifically predicted that after 2040, the phenomenon would no longer be observable. Only two articles referred to the morning observations conducted by Yatsurugi shrine.

Focal point of Japanese articles

The analysis of Japanese articles revealed a significant contrast between their content and that of foreign articles. The primary focus of the Japanese articles was the morning observation conducted by Yatsurugi shrine. While national media covered the event in a single article within the local news category, local media presented the morning observation as a series of features. Notably, one local outlet created a special section titled "*Omiwatari* Watching."

Most national and local media outlets did not mention climate change at all, and none referred to the Industrial Revolution (see Table 1). Instead, the Japanese articles primarily focused on the freezing of Lake Suwa and the actions of the priest and parishioners during the morning observation. Very few articles discussed Shinto legends or the ice ridges.

The national media published one-off articles that were shorter than those in international media, with many being less than half their length. In comparison, local media articles were even more concise, often consisting of just a few lines that reported temperatures from the observation site along with the priest's comments on the ice conditions. While the national media referenced historical records, local media significantly restricted their references.

Observation of the local journalists

Local journalists were observed at the *omiwatari* observation site for three consecutive days each January from 2021 to 2023. Several newspaper companies assigned journalists to the site on a rotating basis, while many attended voluntarily. The journalists' experience with the *omiwatari* observation ranged from one to over thirty years. They gathered at the site even when the likelihood of witnessing the *omiwatari* phenomenon was low due to warm weather or rain.

A friendly atmosphere was evident between the journalists and the shrine members. The journalists often assisted with the observation activities; for example, one journalist used a large flashlight to illuminate the narrow lakeside path for the observers. Although no formal leader emerged, a journalist with long-standing experience appeared to guide the group. This journalist, referred to as the "leader," once told to the author that the primary motivation for reporting on the *omiwatari* was to support the priest and his team in continuing the morning observation tradition. The leader also quietly managed the process to ensure safety, particularly during the COVID-19 pandemic in 2022. In response to the increased alert level for infection control, the leader suggested limiting the number of journalists at the site, and the others followed these instructions.

In contrast, a middle-aged journalist, newly transferred to the Suwa branch, admitted that he was unfamiliar with the *omiwatari* before relocating. He visited a local museum to learn about its history and significance. Although he had only been in Suwa for less

than a year, he recognized the readers' strong interest in the *omiwatari* and the state of Lake Suwa's ice formation during this season.

Journalists at the site operated independently, focusing on various aspects of reporting. Some followed the shrine's parishioners to the lake, while others interviewed visitors at the observation site, asking about their motivations and impressions. This fieldwork identified a few individuals who regularly participated in the morning observation. These individuals were often seen conversing with members of the Yatsurugi shrine community, sharing their hopes for the appearance of the sacred *omiwatari*.

The morning observation concluded with an on-the-spot interview with the priest, who shared his views on the ice formation, connecting it with the divine forces of nature in line with Shinto beliefs. Occasionally, journalists asked about the priest's perspective on warming winters and the continued occurrence of *ake-no-umi*. The priest's responses adhered to a religious interpretation, mentioning that climate is beyond human control, without referencing climate change or global warming during the research period.

To better understand the key themes highlighted by journalists, the narrative in Japanese news articles reporting on the *omiwatari* during the morning observation period in 2022 was analyzed. Additionally, the number of articles covering local events related to the *omiwatari* was counted (Fig. 3).

In 2022, the first ice formation on Lake Suwa was observed on January 3rd. Even though it was outside of the official observation period, two local newspapers mentioned it in conjunction with the *omiwatari*. Subsequently, six of the seven newspapers reported on January 5th, or the following day, that the shrine had initiated its annual observation, aligning with the respective newspapers' publishing schedules. On January 7th, the first complete freezing of the ice was observed, but the ice melted shortly afterwards, only to refreeze fully on January 10th. This cycle of freezing and melting continued. On January 24th, strong winds shattered the lake's ice into large pieces, which washed ashore, creating a visually striking scene. Both locals and tourists gathered to photograph the event, walking on the ice. To prevent accidents, local authorities deployed a drone to warn individuals against approaching the lake and announced that walking on the ice would be considered disrespectful to the divine. By January 29th, the shine's priest announced that the formation of the *omiwatari* was unlikely this year, and the observation officially concluded on February 5th. The shrine conducted a ritual on February 20th to declare *ake-no umi* to the gods, as well as the Meteorological Office and the Imperial Household Agency.

The number of news articles covering the *omiwatari* increased on January 7th, 29th, and February 5th. These dates coincided with significant activities by the shrine, namely the initiation of the annual observation, the priest's statement predicting the absence of *omiwatari*, and the termination of the observation (see Fig. 3).

Focus group interview

Six journalists participated in the focus group interview. All had less than ten years of experience with the *omiwatari* observation. Of the six, three worked for the local branches of major newspaper companies, two for local newspaper outlets, and one for a public broadcast. Two major newspapers prioritized climate change coverage highly, while a national newspaper assigned it moderate importance. In contrast, two local

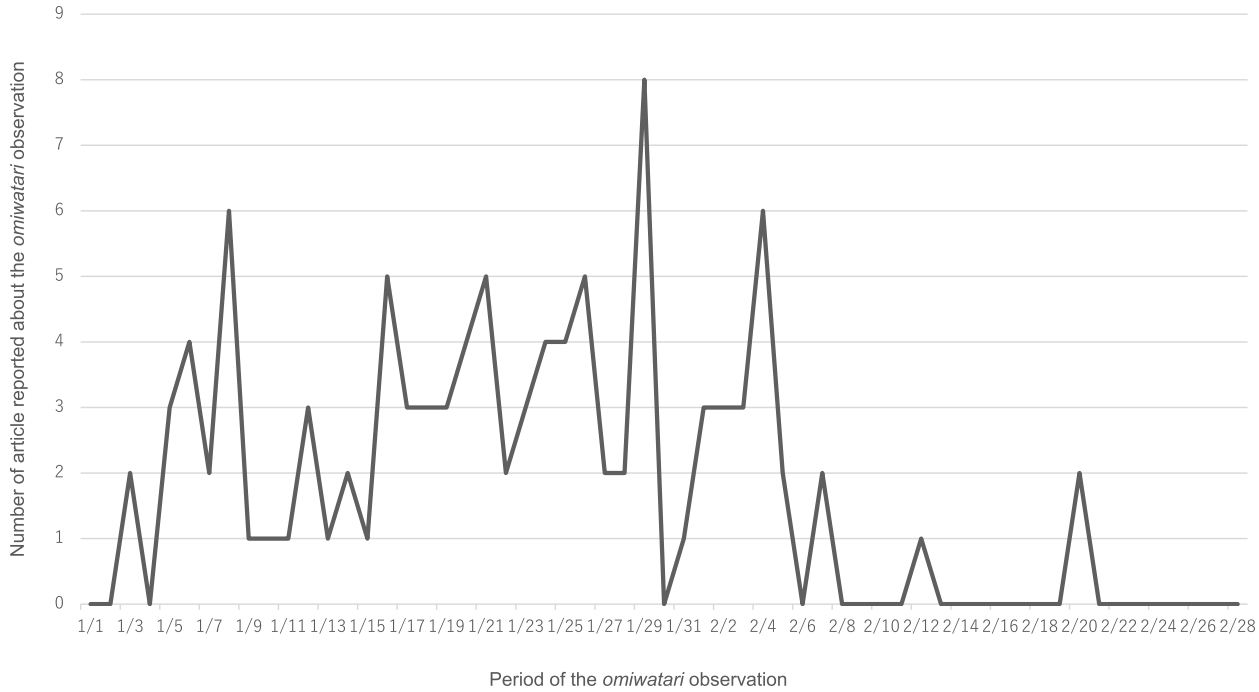


Fig. 3 The number of local media coverage of the annual *omiwatari* observation in 2022. The figure illustrates variations in local media reporting volume, correlating with key events and changes in ice conditions during the observation period

newspapers reported a low priority on climate change, though one journalist anticipated this would rise in the future.

All the participants reported attending the observation site almost daily, driven by the media's responsibility to capture any newsworthy event. One found observing changes in the lake both interesting and a daily routine. All participants agreed that the priest's observations were newsworthy and met the expectations of local readers. One participant aimed to highlight the decline of a seasonal tradition, while another, raised in the prefecture where Lake Suwa is located, emphasized the cultural importance of sharing stories linked to the local myth and honoring ancestral heritage.

Two of the six participants had prior experience reporting on the *omiwatari* phenomenon in relation to climate change. One worked for a national newspaper with a strong focus on climate change, while the other worked for a local media outlet where climate change coverage was moderately prioritized. This local journalist, being from Nagano where the *omiwatari* is more common, had a deeper connection to the phenomenon. The remaining participants had not reported on the *omiwatari* in relation to climate change, despite their organizations' relatively high interest in the topic. One participant explained that they had limited opportunities to report on climate change under their circumstances.

Discussion

This study analyzed the media's portrayal of the *omiwatari* ice tradition, a cultural practice that has been nurtured for five centuries in Suwa, Japan. With warming winters, the ritual—centered on the formation of ice ridges on Lake Suwa, has become increasingly rare. Consequently, the *omiwatari* tradition faces the threat of loss, raising critical questions about how media frame the risks associated with this decline, particularly in relation to climate change. Surprisingly, local media have largely avoided framing diminishing ice ridges, long symbolic of the *omiwatari* tradition, within the context of climte risk. This section discusses how framing by international and domestic media, shaped by journalistic norms, influences efforts to mitigate climate change and preserve this winter cultural heritage.

Thematic framing in international media vs. episodic framing in Japanese media

The comparative analysis from both Japanese and foreign media articles shown in Table 1 revealed significant differences in how the *omiwatari* is defined and framed in each medium. International media primarily employed emphasized the historical and symbolic significance of the ice ridges, often linking the traditional practice—consisting of a Shinto ritual and the recording of the ice—to broader narratives such as the Industrial Revolution and changing winter patterns. In contrast, Japanese media focused on the daily morning observation of the lake ice, centering on the observation group composed of the priest and parishioners of Yatsurugi shrine. The content analysis of each article clearly illustrate the difference in how international and Japanese media frame the issues currently facing the *omiwatari* (Table 1).

Drawing on Iyengar's (1994) framing theory, it is clear that overseas media employed thematic framing, situating the *omiwatari* within the broader context of long-term climate change. This approach highlights the systemic factors contributing to climate

change and the resulting threat to the continuity of the long tradition of the *omiwatari*. On the other hand, Japanese domestic media adopted episodic framing, presenting the issue as an immediate struggle by the people of Yatsurugi shrine to preserve the annual *omiwatari*-related practices without ice ridges. While thematic framing connects the phenomenon to overarching social systems and long-term risks, episodic framing focuses on localized, observable responses of the shrine people to the impacts of changing winter conditions, without reflecting deeply into the underlying causes.

Articles that use thematic framing for climate change often have a short lifespan in public discourse (Sampei & Aoyagi-Usui 2009). This can be a limitation of climate change reporting. To address this gap, overseas media sustain public interest in climate change by incorporating diverse examples of climate risks, such as floods and droughts (Schäfer & Painter 2021). The *omiwatari* tradition represents one such example. By including parallels from Europe and the United States, overseas media contextualize the impact of climate change on the traditions of an island nation like Japan within their broader narrative.

In contrast, Japanese domestic media primarily focus on whether the *omiwatari* phenomenon will occur in the current year. This emphasis is evident from the extensive reporting on the morning observations (Table 1). A striking contrast between the two media lies in their treatment of long-term ice records: while overseas media display significant interest in the historical data, domestic media show minimal engagement with these records, seldom referencing long-term history or future predictions. Even when national media link the morning observations to climate change, the coverage is primarily targeted at local audiences, framing the phenomenon as a seasonal topic rather than a climate change risk. This episodic framing by Japanese media makes it challenging to construct a cohesive narrative around climate change. The interview with local journalists support this discussion, as one of them noted a lack of opportunities to report on climate change in the context of their *omiwatari* coverage.

Content analysis of Japanese media revealed notable differences between national and local outlets. National media reported on the *omiwatari* only when the probability of the phenomenon's appearance was highest or when it was no longer possible. In contrast, local media continuously reported on the possibility of its appearance throughout the lake ice observation period (Fig. 3). National media framed the *omiwatari* as an episode of regional culture within Japan, while local media constructed a narrative specific to the Suwa region, portraying the community's shared interest for the reappearance of the phenomenon as a cherished winter tradition. The Suwa region's focus is evident in the frequent use of the term "expectation" by local media (Table 1).

Journalistic norms shaping the *omiwatari* framing

These differences can be partly explained by the journalistic norms (Boykoff & Boykoff 2007; van Eck et al. 2019) that influence reporting. This study highlighted how norms such as personalization and the need for authoritative sources shape the media strategies used to report the *omiwatari* tradition exposed to climate risks.

Media coverage of climate change is often influenced by cultural contexts and journalistic norms (Boykoff & Boykoff 2007; Lehotský et al. 2019). In this study, overseas media, targeting an international audience, framed the climate risk to the continuity of

the *omiwatari* tradition as novel and dramatic. This provided strong news value, as it emphasized the potential loss of a winter cultural heritage due to climate change. By leveraging these norms, overseas media heightened the appeal of climate change mitigation for global audiences. For instance, five articles predicted that the ice cracks would never appear again, underscoring the severity of the issue.

In contrast, Japanese local media, which placed their articles in local news, prioritized the personalization norm. They focused on the people of Yatsurugi shrine—the custodians of the *omiwatari* tradition—to construct a narrative of the event as a local winter tradition, rather than emphasizing the presence or absence of the ice ridges. Local media particularly centered their reports on visible, everyday changes, such as ice conditions at Lake Suwa (Table 1).

Personalization in reporting can make distant issues like climate change feel more immediate and relevant to the audience (Boykoff & Boykoff 2007; van Eck et al. 2019). However, since Japanese media do not frame the *omiwatari* tradition within the context of climate change, the personalization norm proves ineffective in addressing climate risks. The emphasis on personalization may explain why local Japanese media rarely report climate change as a key factor affecting the omiwatari tradition.

Another complicating factor is the authority-order norm, which requires journalists to rely on credible sources (Anderson 2017). Overseas media incorporated scientific data on climate change and its impact on snow and ice. As shown in Table 1, while these sources heavily relied on Magnuson's research, they successfully linked future climate projections to the declining likelihood of the *omiwatari* phenomenon. In contrast, Japanese national and local media relied on air and water temperatures released by Yatsurugi shrine to predict whether the *omiwatari* ice ridges would occur during the winter (Table 1).

The hesitation of Japanese media to connect the *omiwatari* tradition with climate change may stem from a lack of scientific evidence predicting the future of the phenomenon and its associated rituals. The IPCC report acknowledges the significant cultural losses in the cryosphere that climate change can cause (IPCC 2019). However, research on the climate change risks facing traditional events like the *omiwatari* remains insufficient in Japan (Ministry of the Environment 2020). While some Japanese media mentioned scientific studies on the freezing of Lake Suwa, these references were far fewer than those in foreign media (Table 1). Additionally, no explicit statements linking the absence of ice ridges to climate change were made by the priest—considered an expert on the *omiwatari* tradition—during the study period. Without concrete evidence, journalists adhering to the authority-order norm are unlikely to frame the continuous absense of the *omiwatari* phenomenon within a climate change narrative or address the tradition's future sustainability.

Climate risk management and adaptation in local media coverage of the *omiwatari*

This study found that foreign media's thematic framing of the *omiwatari* tradition highlighted climate risks to encourage climate change mitigation. In contrast, Japanese local media used episodic framing, focusing on the actions of shrine priest and parishioners rather than climate change. This approach may offer a pathway to adapt

the *omiwatari* tradition to climate change by fostering a sense of continuity and familiarity, even in the absence of ice ridges.

Intangible cultural heritage is dynamic, continuously re-created by communities in response to their environment, natural relationships, and historical context (Blake 2008). The *omiwatari* tradition, associated with Yatsurugi shrine, had undergone transformations in the past. The first occurred after the abolition of Japan's feudal system, when the shrine's obligation to report to the national authorities ended, threatening the tradition's legitimacy. With the support of Suwa scientist Fujiwhara, the tradition shifted to a form of citizen science, focusing on recording ice observations. The second transformation involved the introduction of the daily morning observations of Lake Suwa each winter, initiated by the current priest to sustain the tradition. Local media's episodic framing presents the morning observation as a recurring winter event, allowing the tradition to be reported even when the ice ridges—the core symbol—are absent. Through such coverage, readers can perceive the tradition as continuing through the practices maintained by Yatsurugi shrine, reinforcing its cultural relevance.

Adger et al. (2008) argue that the limits of social adaptation to climate change depend on values, perspectives, and worldviews, which determine what changes are perceived as acceptable. Local media's role in framing the omiwatari tradition around practices like the morning observation illustrates how cultural heritage can evolve within these constraints. By redefining the tradition in this way, the community not only preserves its relevance but also creates a narrative that aligns with audience expectations and preferences. Over time, consistent reporting may help audiences reinterpret the tradition as folklore centered on observing and documenting the lake's state. If successful, the episodic framing by local media could play a crucial role in adapting the *omiwatari* tradition to the challenges posed by climate change.

In addition to these constraints, local media in Japan may avoid framing the tradition in terms of climate risk to avoid alienating their audience. Reporting on climate risks associated with a deeply embedded cultural practice could be seen as an intrusion by outsiders into a sacred cultural space (Wanda George 2010) particularly in a community where the tradition has been valued for centuries. As local media must cater to their readers' preferences, de-emphasizing climate change may serve as a risk management strategy in an environment where newspaper subscriptions are already declining (Villi & Hayashi 2017). The positive relationship between journalists and the community of Yatsurugi Shrine, as observed during the fieldwork, may further discourage coverage of climate change in relation to the *omiwatari* tradition.

The efforts by Yatsurugi shrine and local media can also contribute significantly to fostering social capital within the Suwa community. The morning observation and its consistent media coverage may reinforce a sense of shared purpose and cultural significance, uniting residents around a practice deeply embedded in their identity. This aligns with Blake's (2008) view of intangible cultural heritage as a dynamic process that builds community resilience. Furthermore, by redefining the *omiwatari* tradition around the lake observations, the community can ensure continuity while adapting to climatic changes, strengthening their collective ability to navigate challenges and maintain cultural cohesion. The evolving role of the *omiwatari* tradition thus

highlights the interplay between cultural heritage and social capital, underscoring the importance of preserving such traditions for the well-being of local communities.

One local journalist, despite working for a newspaper that does not prioritize climate change coverage, has reported on the *omiwatari* tradition in connection with climate risks. His birthplace in Nagano, where Lake Suwa is located, may have influenced his reporting, as he shared a personal connection with the tradition and the community's values. This example suggests that journalists with strong local ties may be more likely to frame cultural heritage in terms of climate change, as they act both as reporters and stakeholders in the tradition.

Conclusion

This study examined the role of the media in framing a Japanese winter tradition known as *omiwatari* in Nagano, which is currently endangered due to climate change. By analyzing both domestic and international media coverage of this tradition, the study revealed contrasting framing approaches influenced by journalistic norms. International media typically employed thematic framing, presenting the *omiwatari* as a unique example within the broader context of climate change. This approach highlighted systemic risks and aimed to raise awareness about the need for climate change mitigation. In contrast, Japanese local media utilized episodic framing, focusing on the actions of the Yatsurugi shrine priest and parishioners who are working to preserve the tradition—often without explicitly mentioning climate change. The findings highlight the important role of local media in balancing the preservation of cultural heritage with the need to address climate risks. While the omission of climate-related framing may be an effort to respect cultural sensitivities and engage local audiences, it can also limit public awareness of the systemic threats posed by climate change. Nevertheless, the episodic framing used by local media, which centers on the daily observations of the current priest, offers an alternative pathway for adapting the *omiwatari* tradition to the impacts of climate change. By reinterpreting the tradition as a practice centered on lake observation, local media can help maintain its cultural relevance and community interest, even in the absence of ice ridges. In conclusion, the media's framing of the *omiwatari* has significant implications for preserving winter cultural heritage in the face of climate change. A nuanced approach that respects local traditions while also paying greater attention to climate risks could enhance both cultural sustainability and public understanding of environmental challenges. Future research should investigate the long-term impact of these framing strategies on the sustainability of the *omiwatari* tradition and explore the community's perspectives on the preservation of winter cultural heritage and climate adaptation.

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Authors' contributions
The author solely contributed to the conception, design, data collection, analysis, and writing of this manuscript.

Data availability
This study used media articles and interview transcriptions, which constitute datasets. These datasets can be made available upon request..

Declarations

Ethics approval and consent to participate
Ethics approval and consent to participate are not applicable to this study.

Competing interests
This research was financially supported by JST (Japan Science and Technology Agency). The author declare that they have no competing financial or non-financial interests related to the content of this article.

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